Warehouse Management (WM) I

This case study explains an integrated warehouse management process which is triggered by a purchase order for a warehouse-managed storage location.

Product

S/4HANA 2023
Global Bike

Fiori 3.0

Level

Beginner

Focus

Warehouse Management

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Version

4.3

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MOTIVATION

Warehousing has significant value for logistics.

Current trends such as higher cost pressure, shorter cycles of innovation, higher customer expectations and globalization of markets make great demands on companies, particularly on warehouse logistics. This is especially difficult in industries with high differentiation like the consumer goods industry. Furthermore, customers have increasingly higher demands on reliability, promptness and flexibility of deliveries.

Warehouse management systems support the global flow of goods between the producer and the purchaser and facilitate near fail-proof logistic operations in increasingly complex supply chains.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to work successfully through this case study, it is not necessary to finish the WM exercises. Anyway, it is recommended.

NOTES

This case study uses the Global Bike (GB) data set, which has been exclusively created for SAP UA global curricula.

SAP University Alliances

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Process Overview

Learning Objective Understand and perform a warehousing process for externally procured goods. **Time** 65 min

Scenario Due to increasing sales output in your San Diego distribution center, management has decided to install a Warehouse Management System. This implementation has just been completed and the new system needs to be tested. For this purpose, trading goods should be procured by a vendor and put in the stock in San Diego, using the new warehouse management system.

Employees involved Jennifer Brown (Plant Manager)
Carolin Bruzik (Warehouse Supervisor)
Sunil Gupta (Warehouse Employee)
Yoshi Agawa (Goods Receipt Clerk)

In order to receive goods from a vendor you need to create a purchase order. Goods will be sent by the vendor to the distribution center and you will create a goods receipt in San Diego. The system will automatically create a transfer order for the received goods to put them into stock. In conclusion, you will check if the goods were stored in the correct storage bins. As this case study focuses on Warehouse Management, detailed instructions of how to receive the invoice and how to pay the vendor are not included. However, you may use respective parts of the Materials Management (MM) case study to finalize the procurement process and see the financial impact.

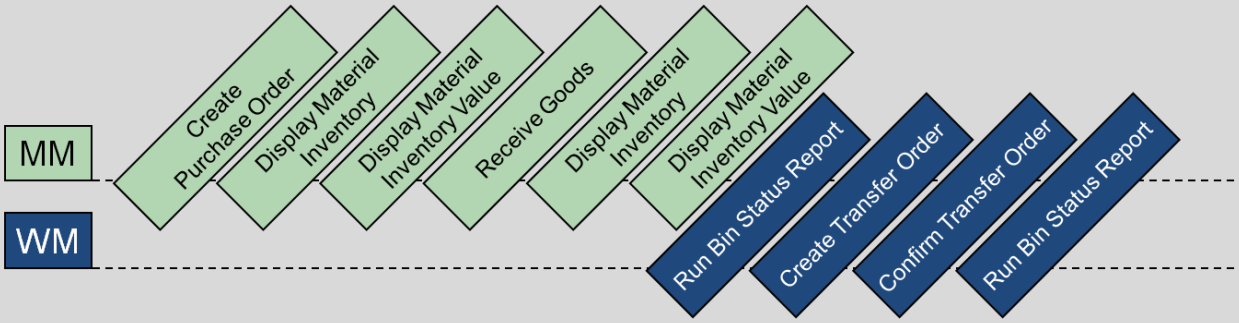


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Step 1: Create Purchase Order

Task Create a purchase order.

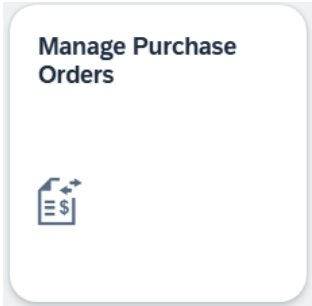
Time 10 min

Short Description Use the SAP Fiori Launchpad to create an immediate purchase order for materials from a supplier, i.e. to start the procurement process without having created a purchase requisition before.

Name (Position) Jennifer Brown (Plant Manager)

To create a purchase order, use the app *Manage Purchase Orders* in the *Warehouse Management* area on the *Storage Purchasing* page in the *Plant Manager* role.

Start



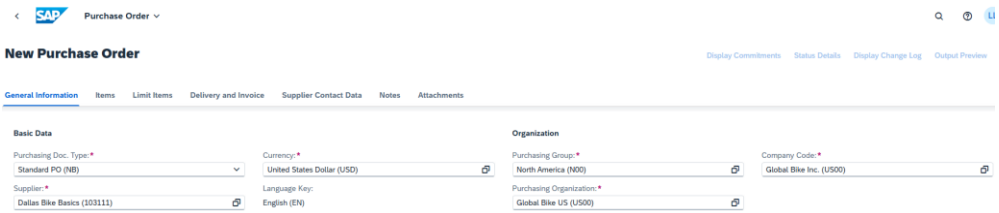
Click on Create.

In the *Purchase Order* screen, enter the *type of purchasing Doc*. **NB Standard PO** and **103###** (replace ### with your number) as *Supplier*.

NB Standard PO
103###

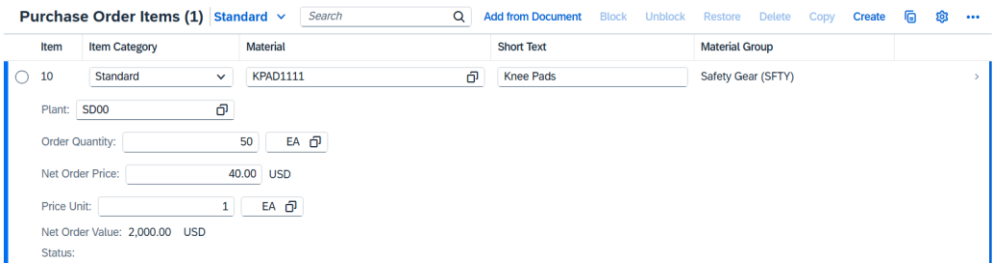
Then, fill in **N00** as *Purchasing Group*.

N00



In the *Items* tab, click on Create to create the item. Enter **KPAD1###** as the *material* (replace ### with your number), **SD00** as the *plant*, **50** as the *order Quantity*, **40** USD as the *net order price*.

KPAD1###
SD00
50
40



Click on › to go to the *Purchase Order Item* screen.

In the *Delivery details* tab, enter **TG00** as the *storage location*. In the *Schedule lines* tab, enter **8 days from today** as the *delivery date*.

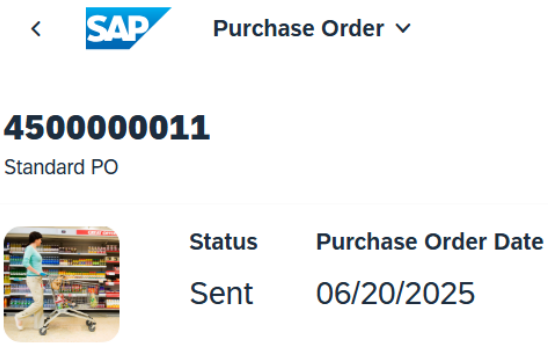
TG00
8 days from today

Click on Apply.

In the *Items tab*, click again on Create, but select **EPAD1###** as the *material* and repeat your entries.

Repeat for EPAD1###

Click on Apply. Then click on Order to save your order. The system assigns a document number for the standard order.



Click on SAP to return to the SAP Fiori Launchpad.

Step 2: Display Material Inventory

Task View the inventory of your material.

Time 5 min

Short Description Use the SAP Fiori Launchpad to display the inventory of your material.

Name (Position) Jennifer Brown (Plant Manager)

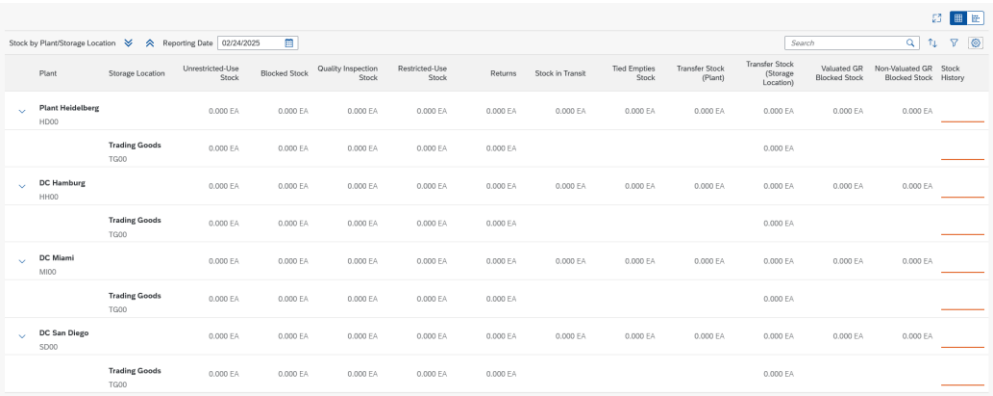
To display the material stock, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Plant Manager* role the app *Stock - Single material*.

Start

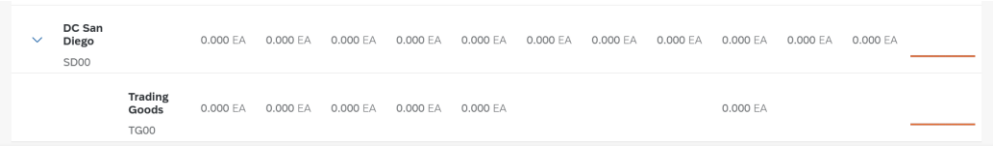


Enter **KPAD1###** as Material (remember to replace ### with your number). Click on Enter.

KPAD1###

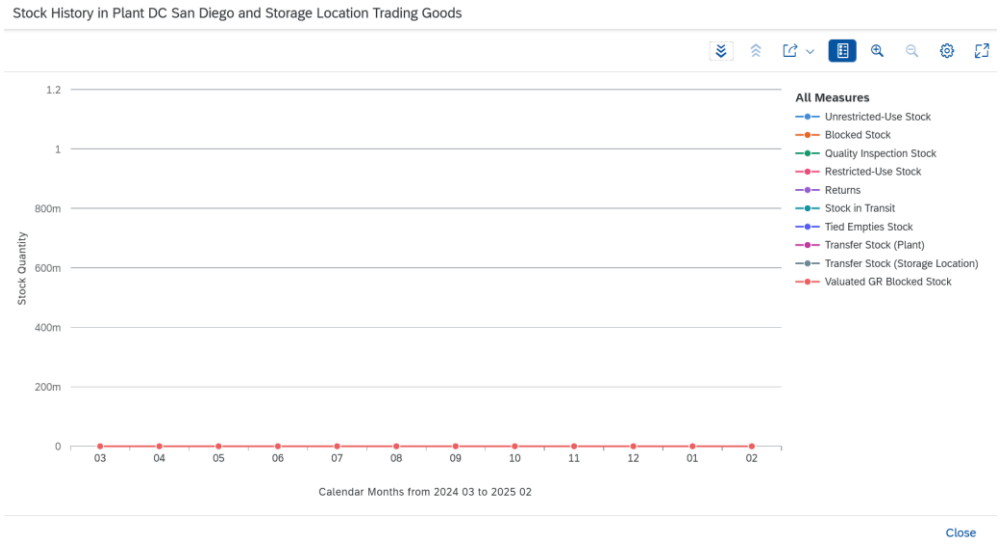


Note that the unrestricted-use stock of goods in San Diego is zero. If you can not see the *Stock History* you can easily insert it in the *Settings* .



Click on the icon in the row for DC San Diego in the Stock History symbol .

The system will now display the stock history of your material KPAD1### in SD TG00.



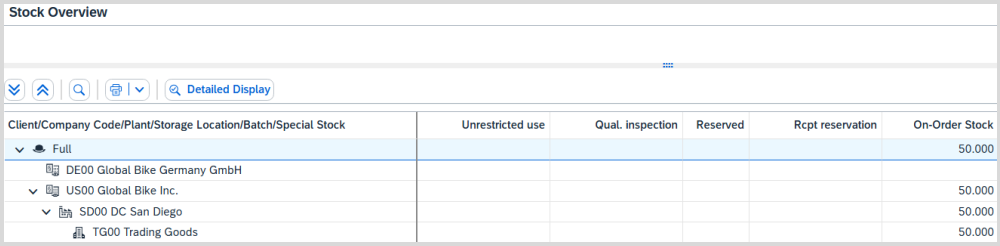
Repeat this task for the material **EPAD1###**.

Repeat for EPAD1###

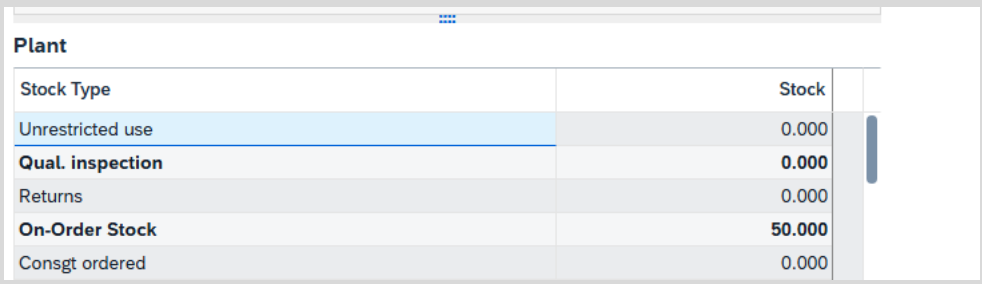
Click on SAP to return to the SAP Fiori Launchpad.

KPAD1###

Note To display the material inventory, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Plant Manager* role the app *Display Stock Overview*. Enter **KPAD1###** as Material



After double clicking on *SD00 DC San Diego*, you will see a separate *Stock Overview* for your distribution center and the On-Order Stock balance of 50.



Click on SAP to return to the SAP Fiori Launchpad.

Step 3: Display Material Inventory Value

Task View the value of your material inventory.

Time 5 min

Short Description Use the SAP Fiori Launchpad to display your material inventory.

Name (Position) Jennifer Brown (Plant Manager)

To display the material inventory value, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Plant Manager* role the app *Display Warehouse Stock*.

Start



In the *Display Warehouse Stocks of Material* screen, enter **KPAD1###** as Material. Ensure that all other search criteria fields are blank and click on Execute.

KPAD1###

Material			Material Description		Plnt Name 1		
Locat SL	Unrestricted Total Value	Unit Crcy	Transit/Transf. Total Value	Quality Inspection Total Value	Restricted-Use Total Value	Blocked Total Value	Returns Total Value
KPAD1100			Knee Pads		HD00 Plant Heidelberg		
TG00	0	EA	0	0	0	0	0
	0.00	EUR	0.00	0.00	0.00	0.00	0.00
KPAD1100			Knee Pads		HH00 DC Hamburg		
TG00	0	EA	0	0	0	0	0
	0.00	EUR	0.00	0.00	0.00	0.00	0.00
KPAD1100			Knee Pads		MI00 DC Miami		
TG00	0	EA	0	0	0	0	0
	0.00	USD	0.00	0.00	0.00	0.00	0.00
KPAD1100			Knee Pads		SD00 DC San Diego		
TG00	0	EA	0	0	0	0	0
	0.00	USD	0.00	0.00	0.00	0.00	0.00

You can see that all values listed for this material are currently zero.

Repeat this task for the material **EPAD1###**.

Repeat for EPAD1###

Click SAP to return to the SAP Fiori Launchpad.

Step 4: Receive Goods

Task Receive goods at receiving plant.

Time 5 min

Short Description Use the Fiori Launchpad to create a goods receipt which documents the receiving of your materials in San Diego.

Name (Position) Yoshi Agawa (Goods Receipt Clerk)

To create a goods receipt, use the app *Post Goods Receipt for Purchasing Document* in the *Warehouse Management* area on the *Storage Purchasing* page in the *Goods Receipt Clerk* role.

Start



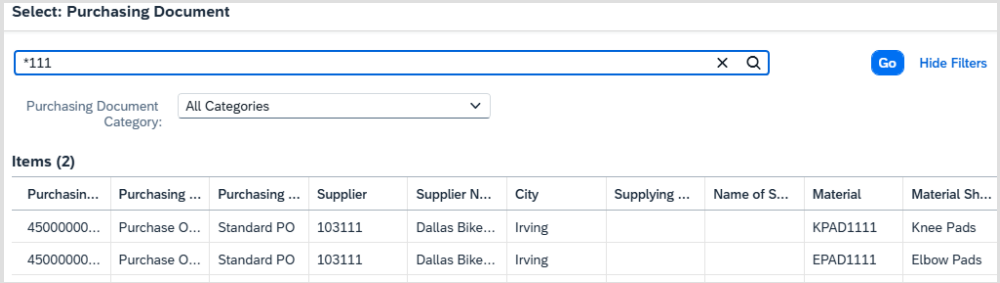
Enter your **purchasing document number** in the *Purchasing document* field.

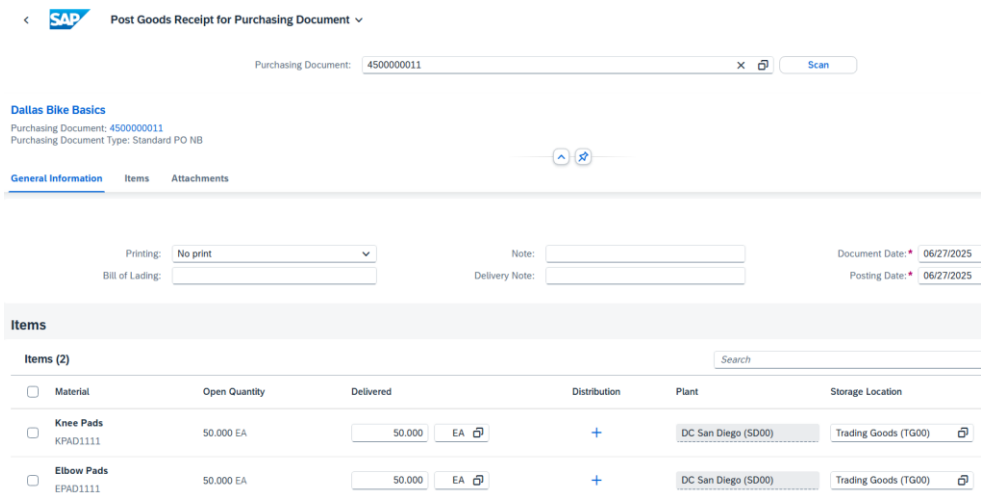
Purchasing Document Number

Note If you want to search for your *purchasing document number*, use the F4 help.

Enter your three-digit number *### in the *Search* field and click on Go. Your purchasing document is displayed. Double-click on it to accept it.

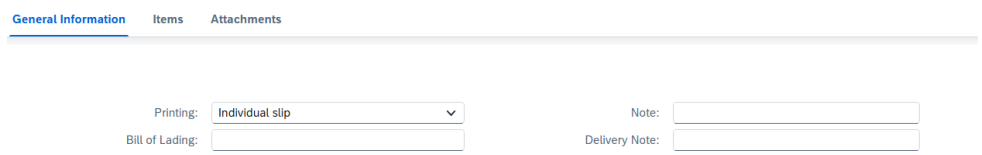
*###





Select **Individual slip** for *printing*.

Individual slip

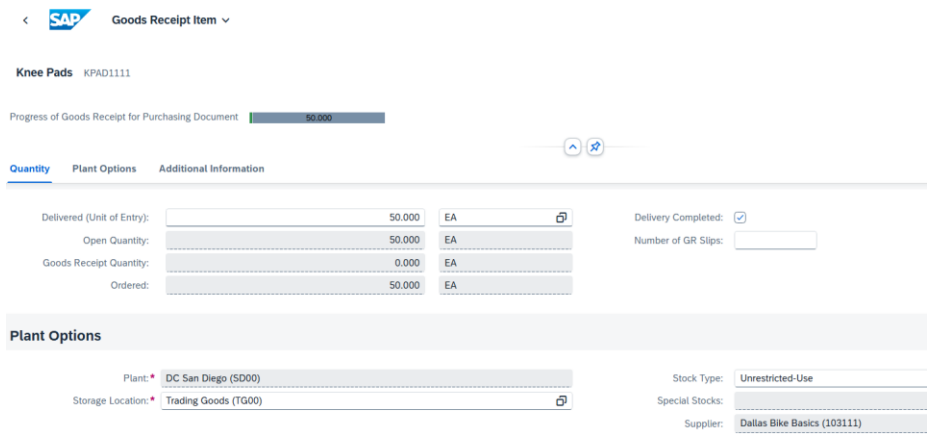


Click on line *Material* KPAD1### to see detailed information.

KPAD1###

Check that the *storage location* **Trading Goods** and *stock type* is set to **Unrestricted-Use**. Select the checkbox for **delivery completed.**

Trading Goods
Unrestricted Use



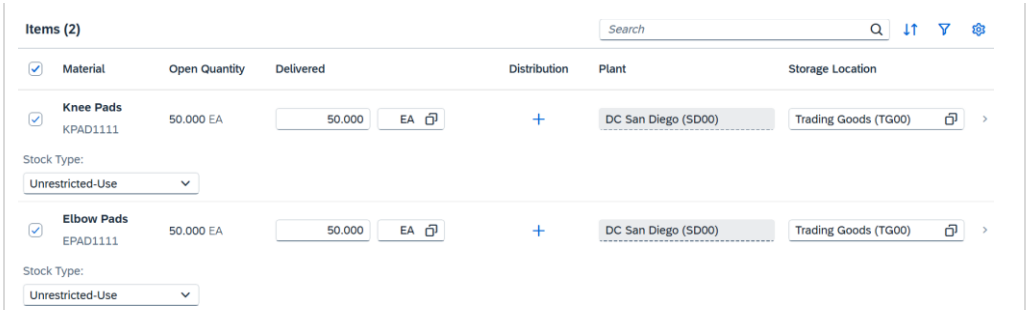
Now click on Apply.

Repeat this step for **EPAD1###**.

EPAD1###

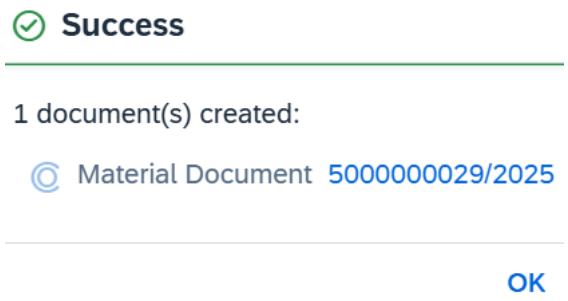
Now click on Apply.

Leave both lines selected.



Then, click on Post to save your receipt. The system will assign a unique material document number.

Material document number



Click on SAP to return to the SAP Fiori Launchpad.

Step 5: Display Material Inventory

Task View the inventory of your material again.

Time 5 min

Short Description Use the Fiori Launchpad to display the inventory of your material again.

Name (Position) Jennifer Brown (Plant Manager)

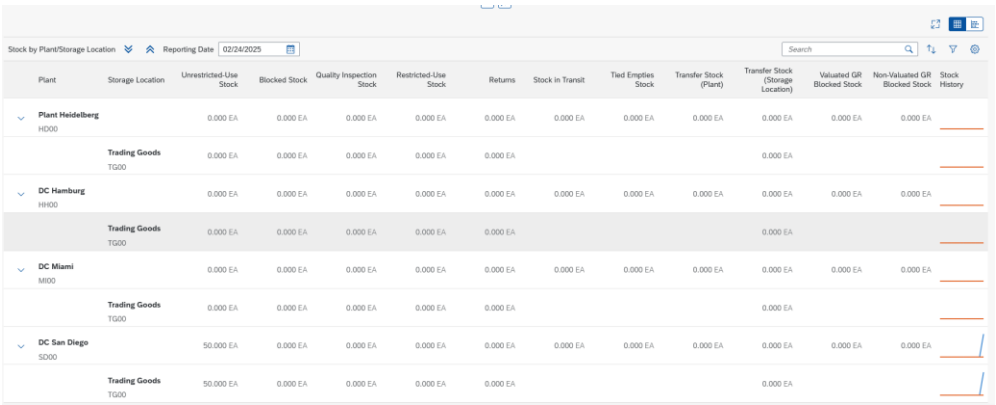
To display the material stock, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Plant Manager* role the app *Stock - Single material*.

Start

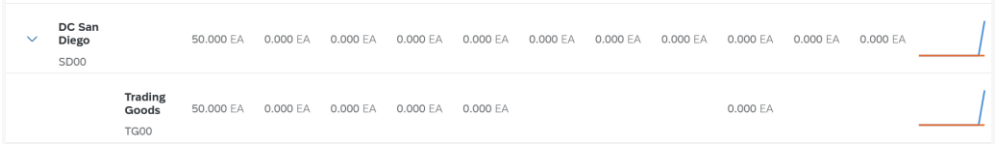


Enter **KPAD1###** as Material (remember to replace ### with your number). Lick on Enter.

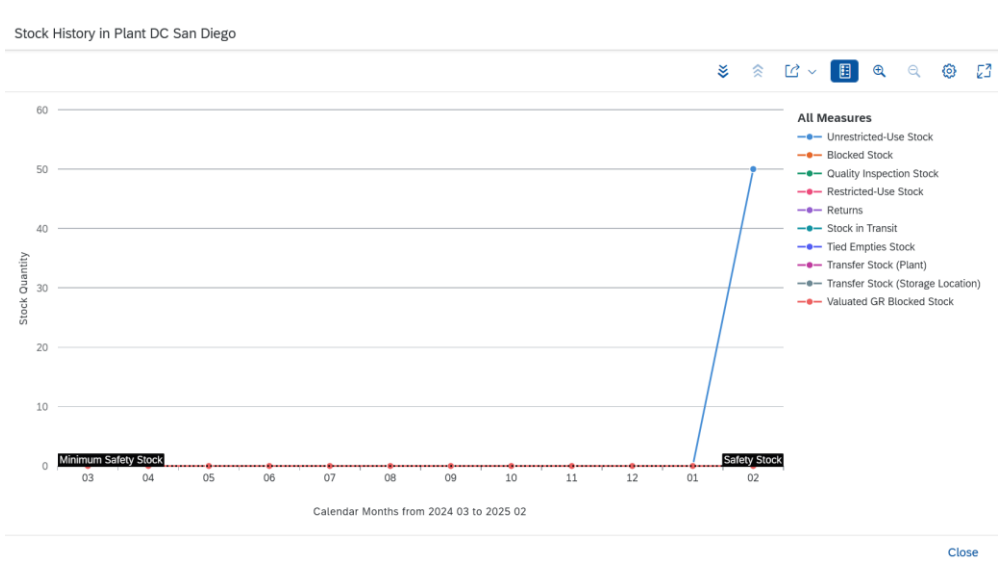
KPAD1###



Note that the unrestricted-use stock of goods in San Diego has changed.



Click on the icon in the row for SD San Diego in the Stock History symbol . The system will now display the stock history of your material KPAD1### in SD TG00.

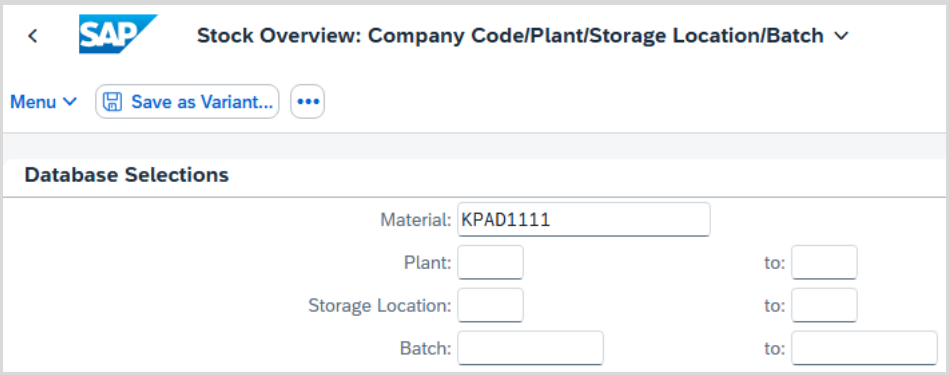


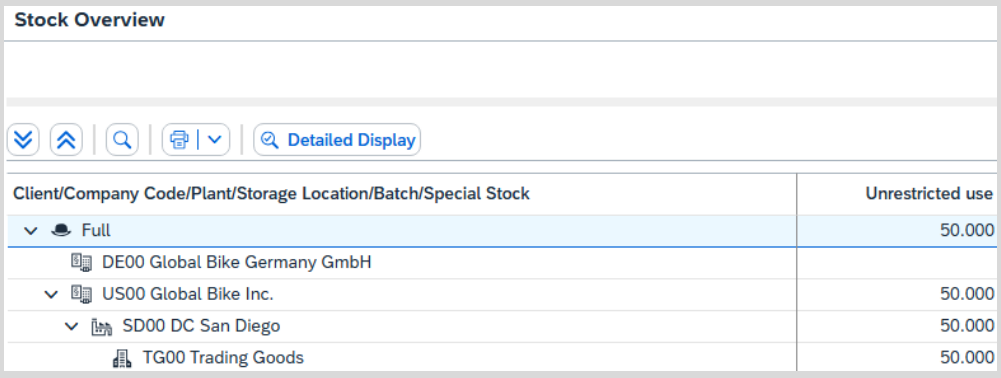
Repeat this task for the material **EPAD1###**.

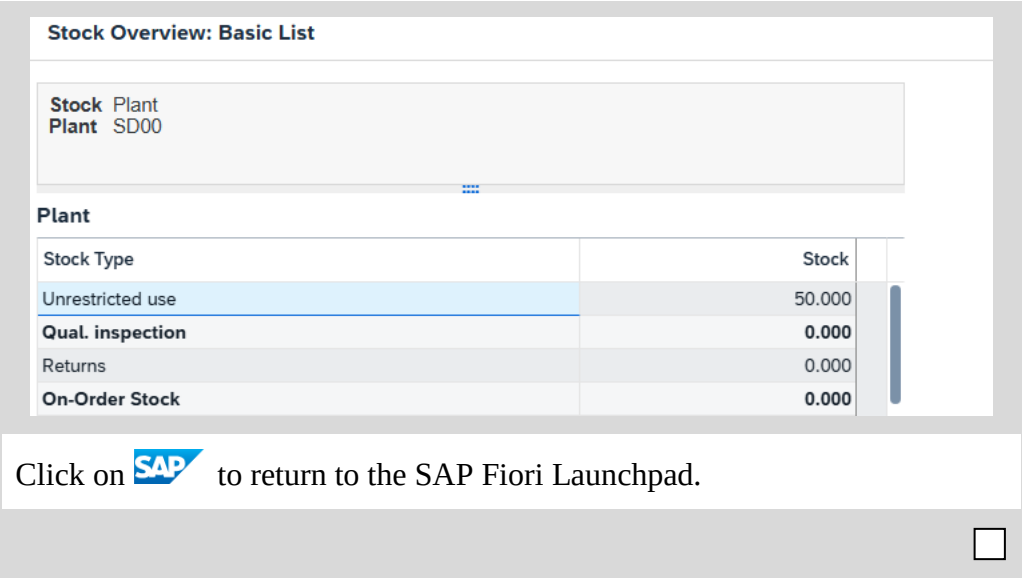
Repeat for EPAD1###

Click on SAP to return to the SAP Fiori Launchpad.

Note To display the material inventory, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Plant Manager* role the app *Display Stock Overview*.







Click on SAP to return to the SAP Fiori Launchpad.

Step 6: Display Material Inventory Value

Task View the value of your material inventory again.

Time 5 min

Short Description In this step, you will use the Fiori Launchpad to display the value of your material inventory again.

Name (Position) Jennifer Brown (Plant Manager)

To display the material inventory value, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Plant Manager* role the app *Display Warehouse Stock*.

Start



In the *Display Warehouse Stocks of Material* screen, enter **KPAD1###** as Material (remember to replace ### with your number). Ensure that all other search criteria fields are blank and click on Execute.

KPAD1###

Material		Material Description		Plnt Name 1		
Locat SL	Unrestricted Unit	Transit/Transf.	Quality Inspection	Restricted-Use	Blocked	Returns
	Total Value Crcy	Total Value	Total Value	Total Value	Total Value	Total Value
KPAD1100		Knee Pads		HD00 Plant Heidelberg		
TG00	0 EA	0	0	0	0	0
	0.00 EUR	0.00	0.00	0.00	0.00	0.00
KPAD1100		Knee Pads		HH00 DC Hamburg		
TG00	0 EA	0	0	0	0	0
	0.00 EUR	0.00	0.00	0.00	0.00	0.00
KPAD1100		Knee Pads		MI00 DC Miami		
TG00	0 EA	0	0	0	0	0
	0.00 USD	0.00	0.00	0.00	0.00	0.00
KPAD1100		Knee Pads		SD00 DC San Diego		
TG00	50 EA	0	0	0	0	0
	2,000.00 USD	0.00	0.00	0.00	0.00	0.00

As you can see the value for the 50 units of your material has been added to the distribution center in San Diego.

Repeat this task for the material **EPAD1###**.

EPAD1###

Click on SAP to return to the SAP Fiori Launchpad..

Step 7: Run Bin Status Report

Task Check the status of your bins.

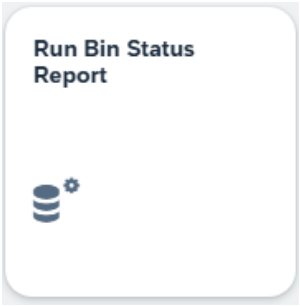
Time 5 min

Short Description Use the SAP Fiori Launchpad to run a bin status report, which will display a detailed report of each storage bin within the specified warehouse.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

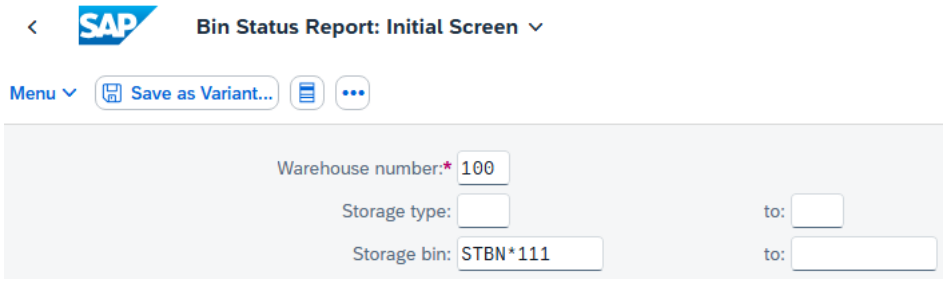
To run a bin status report, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Warehouse Supervisor* role the app *Run Bin Status Report*.

Start

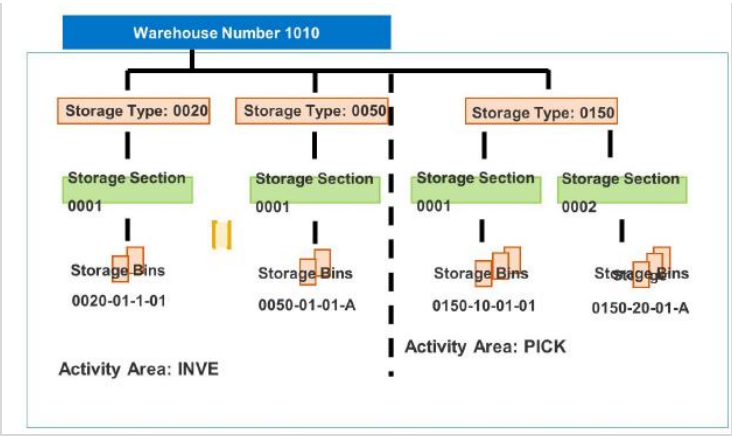


In the *Bin Status Report: Initial Screen*, enter **100** (for your San Diego Warehouse) as Warehouse Number and **STBN*###** as Storage bin (replace ### with your number).

100
STBN*###



Note: The warehouse number is the highest level of organizational unit in warehouse management. In practice, the warehouse number usually corresponds to a physical building or distribution center. Each warehouse number has a substructure that maps the spatial relationship in the warehouse complex in detail.



Storage bins are the lowest level of organizational structure. They are assigned to a storage type and a storage section (if one exists). Storage bins represent the physical location where the goods are stored in the warehouse.

Then, click on Execute.

Typ	StorageBin	Material	Plnt
001	STBN-1-111	<<empty>>	
001	STBN-2-111	<<empty>>	
001	STBN-3-111	<<empty>>	
002	STBN-7-111	<<empty>>	
002	STBN-8-111	PRTR1111	SD00
002	STBN-8-111	PRTR2111	SD00
002	STBN-9-111	ORMN1111	SD00
002	STBN-9-111	ORWN1111	SD00

In the *Bin Status Report: Overview* screen you should see a list of all your storage bins for the entire warehouse in San Diego. Double click on one of your storage bins to get detailed information. As you can see the ordered materials are not present yet. Currently they are located in temporary bins.

Click on SAP to return to the SAP Fiori Launchpad.

Step 8: Create Transfer Order

Task Create a transfer order.

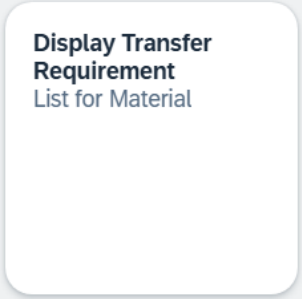
Time 10 min

Short Description Use the Fiori Launchpad to create a transfer order to place your goods into your storage bin. It is a handoff from inventory management to warehouse management. The system recognizes that there are goods that have been received but need to be put away.

Name (Position) Sunil Gupta (Warehouse Employee)

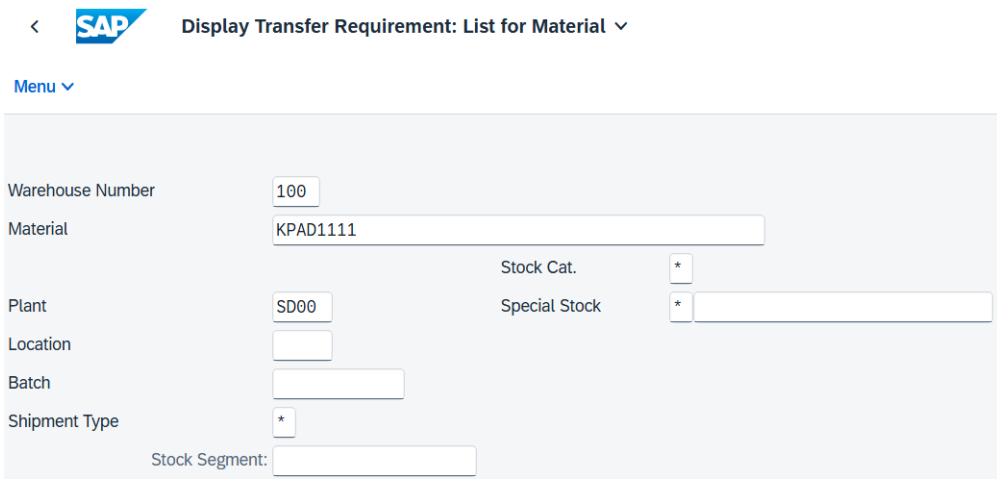
To create a transfer order, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Warehouse Employee* role the app *Display Transfer Requirement – List for Material*.

Start

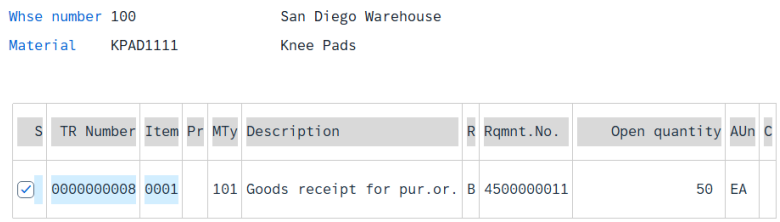


In the *Display Transfer Requirement: List of Material* screen, enter **100** (for your San Diego Warehouse) as Warehouse Number, **KPAD1###** as Material (replace ### with your number) and **SD00** as Plant. Then, press Enter.

100
KPAD1###
SD00



In the *Transfer Requirements for Material* screen, you should see a line item describing the goods just received for your purchase order. The requirement number should be the same as the purchase order number you received earlier. Make sure that the line item is selected.



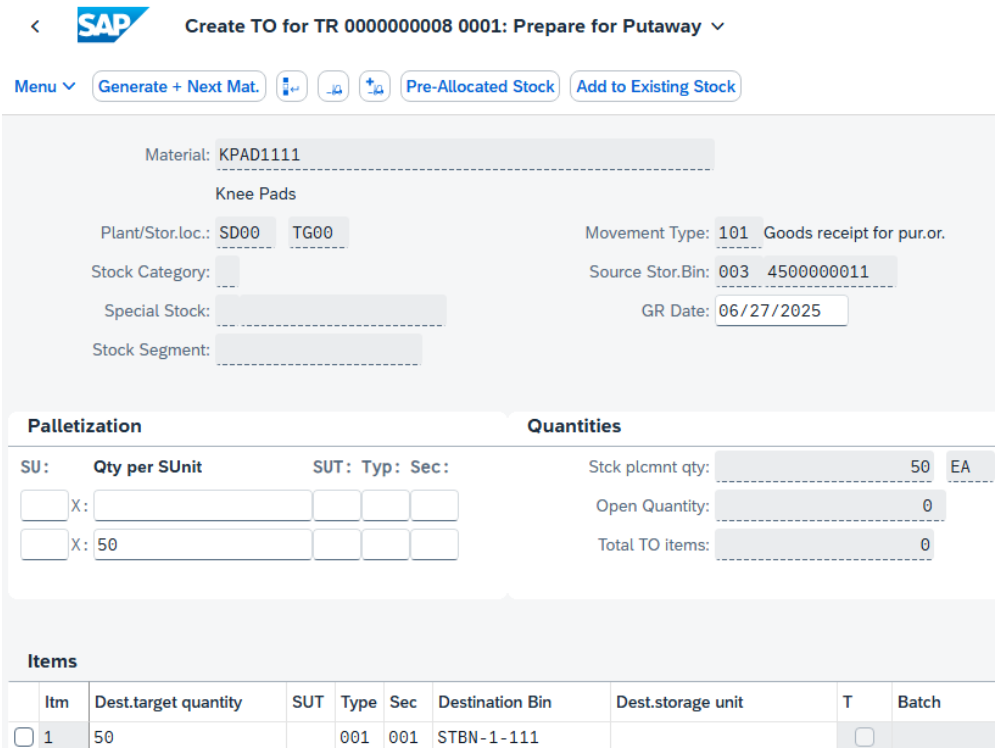
Whse number 100 San Diego Warehouse
Material KPAD1111 Knee Pads

S	TR Number	Item	Pr	MTy	Description	R	Rqmnt.No.	Open quantity	AUn	C
☑	0000000008	0001		101	Goods receipt for pur.or.	B	4500000011	50	EA	

Click on TO in Foreground button.

In the *Create TO for TR 00000000XX 0001: Prepare for Putaway* screen, hit Enter to copy your quantity of 50 from the *Palletization* section to the *Items* section. Enter **001** as *Sec*, **STBN-1-###** as *Destination Bin* (replace ### with your number) and use F4 to select **Shelf Storage** a *Type*. Confirm your entries by pressing Enter.

001
STBN-1-###
001 (Shelf Storage)



Create TO for TR 0000000008 0001: Prepare for Putaway

Menu | Generate + Next Mat. | Pre-Allocated Stock | Add to Existing Stock

Material: KPAD1111
Knee Pads
Plant/Stor.loc.: SD00 TG00
Movement Type: 101 Goods receipt for pur.or.
Stock Category:
Source Stor.Bin: 003 4500000011
Special Stock:
GR Date: 06/27/2025
Stock Segment:

Palletization

SU:	Qty per SUnit	SUT:	Typ:	Sec:
X:				
X:	50			

Quantities

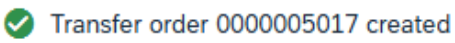
Stck plcmnt qty: 50 EA
Open Quantity: 0
Total TO items: 0

Items

Itm	Dest.target quantity	SUT	Type	Sec	Destination Bin	Dest.storage unit	T	Batch
1	50		001	001	STBN-1-111			

Then, click on Posting to save your transfer order. If any warnings occur ignore them by pressing Enter. The system will assign a unique transfer order number. Please write down this number.

Transfer Order Number



Transfer order 0000005017 created

Repeat the whole procedure for your material **EPAD1###** to put it in the same storage bin.

Transfer Order Number

Click on SAP to return to the SAP Fiori Launchpad.

Step 9: Confirm Transfer Order

Task Confirm your transfer order.

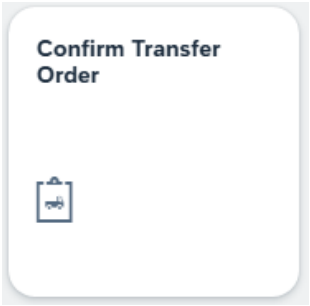
Time 10 min

Short Description Use the Fiori Launchpad to confirm the transfer order you created in the previous step. This is to confirm that the goods are physically in the storage bin indicated in the transfer order.

Name (Position) Sunil Gupta (Warehouse Employee)

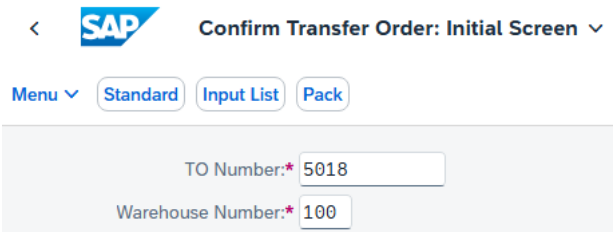
To confirm a transfer order, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Warehouse Employee* role the app *Confirm Transfer Order*.

Start

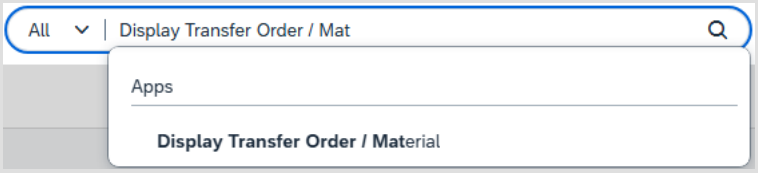


In the *Confirm Transfer Order: Initial Screen*, enter the **Transfer Order Number** from the previous task and **100** as Warehouse Number. Then press Enter.

Transfer Order Number
100



Note If you have not written down the number, you can search for it using the app *Display Transfer Order / Material*.

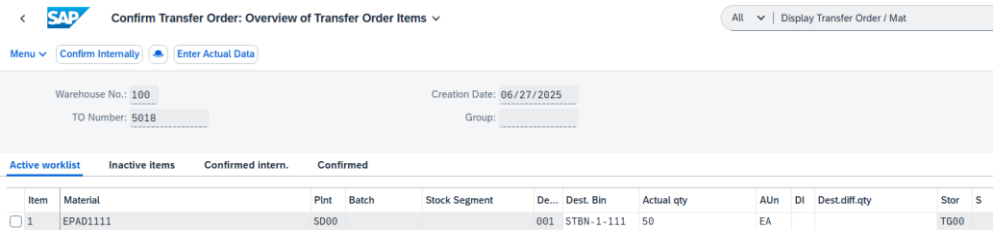


In the *Transfer Orders for Materials* you have to fill in **100** as *Warehouse number* and your material **KPAD1### / EPAD1###** to display the regarding transfer order. Then, click on Execute.

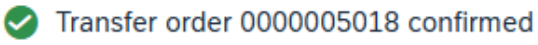
100
KPAD1###
EPAD1###

In the *Confirm Transfer Order: Overview of Transfer Order Items* screen you should see an overview of your transfer order created in the previous step.

Review all of the details to make sure you have the correct quantity and storage bin.



Then, click on Posting to confirm your transfer order. The system will return a success message.



Repeat this step for your other transfer order number (second material).

Repeat for second Transfer Order

Click on SAP to return to the SAP Fiori Launchpad.

Step 10: Run Bin Status Report

Task Check the status of your bins again.

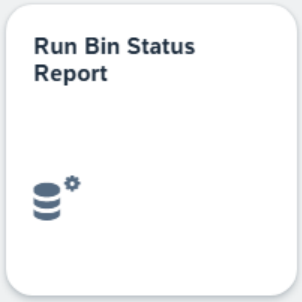
Time 5 min

Short Description Use the Fiori Launchpad to run a bin status report, which will display a detailed report of each storage bin within the specified warehouse.

Name (Position) Carolin Bruzik (Warehouse Supervisor)

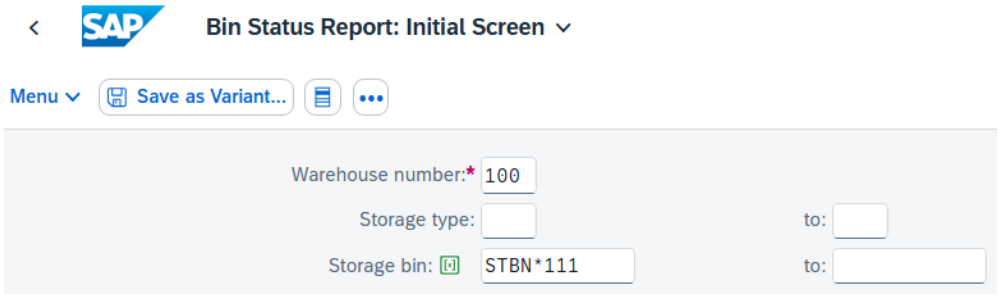
To run a bin status report, use in the *Warehouse Management* area on the *Storage Purchasing* page in the *Warehouse Supervisor* role the app *Run Bin Status Report*.

Start



In the *Bin Status Report: Initial Screen*, enter **100** (for your San Diego Warehouse) as Warehouse Number and **STBN*###** as Storage bin (replace ### with your number).

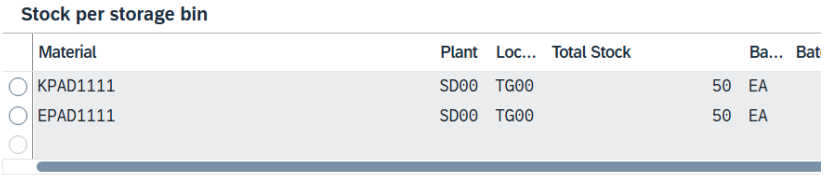
100
STBN*###



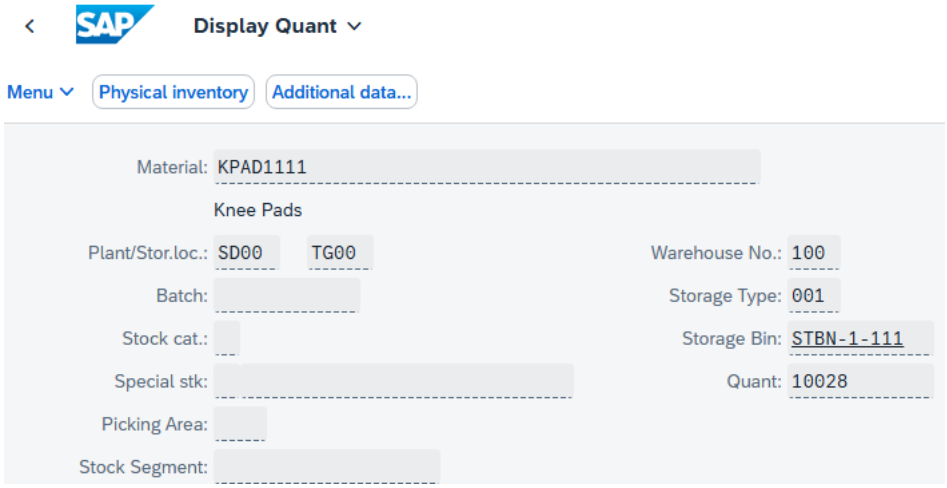
Then, click on Execute.

Typ	StorageBin	Material	Plnt
001	STBN-1-111	EPAD1111	SD00
001	STBN-1-111	KPAD1111	SD00
001	STBN-2-111	<<empty>>	
001	STBN-3-111	<<empty>>	
002	STBN-7-111	<<empty>>	
002	STBN-8-111	PRTR1111	SD00
002	STBN-8-111	PRTR2111	SD00
002	STBN-9-111	ORMN1111	SD00
002	STBN-9-111	ORWN1111	SD00

In the *Bin Status Report: Overview* screen you should see that the Storage Bin **STBN-1-###** is filled now. Click on this storage bin to display detailed information and check whether 50 of each of your goods are stored in it.



You can also display the quant assigned to each material. To do this, select a material and press Quant.



Click on SAP to return to the SAP Fiori Launchpad.

WM I Challenge

Learning Objective Understand and perform a warehousing process for ext. goods.

Time 40 min

Motivation After having finished the *Warehouse Management I* case study successfully, you should now be able to solve the following challenge.

Scenario The warehouse management system has been tested without any problems, so the management decided to use the system productively. Now your task is to order two different products (water bottles and road helmets) from the supplier *Spy Gear*, 50 pieces each. A water bottle will cost 11 USD and a road helmet will cost 27 USD. The trading goods should be delivered in 8 days.

After receipt of the goods in your goods SD00 in San Diego, they are to be stored separately in two different storage bins. The goods are stored in the destination storage bins STBN-2-### for the water bottles and STBN-3-### for the street helmets.

Task Information You can use the *Warehouse Management I* case study as a guideline, but it is recommended to complete this challenge without further assistance to prove your WM skills.